Lexical innovation: new words and meanings

*Lexicology and Lexicography — **Course Website***

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Outline

- **Cultural-linguistic innovation**: The interplay of society and language
- **S-curve model**: Diffusion and change patterns
- **EC-Model**: Entrenchment and conventionalisation
- **Diffusion pathways**: Variation and spread
- **Operationalisation**: Frequency measures and corpus methods
- **Practice**: Studying lexical innovations in corpora
- **Summary**: Key takeaways

From words to innovation

1. **Word formation processes**: Which processes create completely new word forms?
2. **Semantic change**: How can existing words develop new meanings?
3. **Dictionary evidence**: How do dictionaries track and document new words?

Today's focus: How do these processes work in real time? What drives innovation and how does it spread?

Research foundation

Social networks of lexical innovation

Würschinger, Quirin. 2021. "Social Networks of Lexical Innovation. Investigating the Social Dynamics of Diffusion of Neologisms on Twitter." *Frontiers in Artificial Intelligence* 4: 106. https://doi.org/10.3389/frai.2021.648583.

Research approach

- **Longitudinal study**: 99 English neologisms tracked on Twitter
- **Multi-method analysis**: Frequency measures + social network analysis
- **Temporal dynamics**: Usage patterns, volatility, and diffusion over time
- **Social pathways**: How innovations spread through communities

Key insights

- **Frequency alone is insufficient** - need temporal and social information
- **Social diffusion varies significantly** - even words with similar frequency show different patterns

Today’s session structure

We’ll explore the theoretical framework and empirical findings:

1. **Theoretical models**: S-curve and EC-Model for understanding innovation
2. **Operationalisation**: How to measure diffusion using corpus methods
3. **Practical application**: Hands-on analysis with case study words
4. **Comparative analysis**: Your findings vs. published research

Goal: Understand both the theory and practice of studying lexical innovation

The interplay of cultural and linguistic innovation

Society and language change

Key principle: Society continually changes as new practices and products emerge.

- These changes typically **first manifest themselves in language** on the level of lexis.
- Language change occurs through **neologisms** (new words or new meanings).
- **Knowledge of words is conventional**: speakers learn form-meaning pairings.

Types of neologisms

Formal neologisms

- *smartphone*
- *iPhone*
- *Covid*

Semantic neologisms

- *hotspot* (from physical location → infection cluster)
- *social distancing*
- *spreader* (person → virus transmitter)
- *Querdenker* (lateral thinker → conspiracy theorist)

Lexical renaissance

Innovation after lexical attrition: old words returning to use.

Examples:

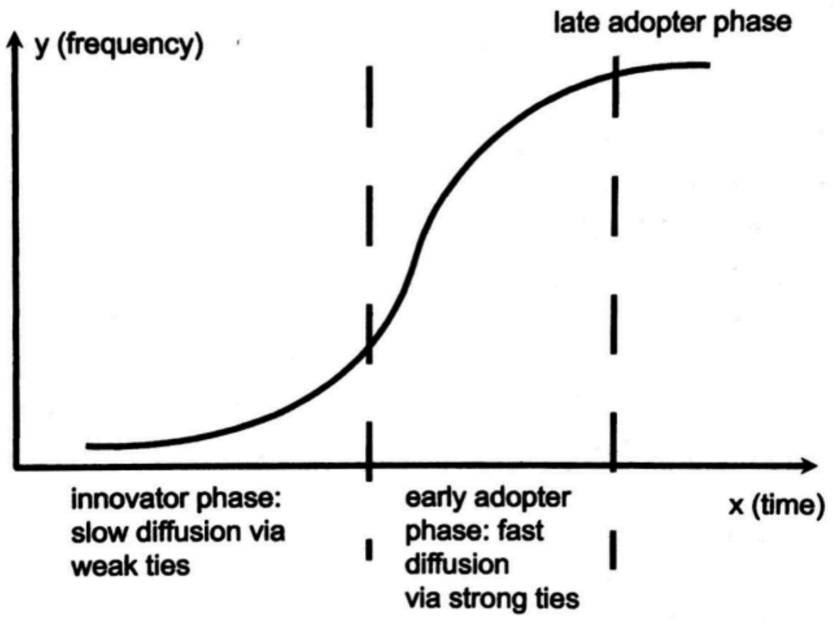
- *dotard* (revived during political discourse)
- *to furlough* (returned during economic crises)

Pattern: Social changes can revive forgotten vocabulary when circumstances make old words newly relevant.

The S-curve model of lexical innovation

Integration of diffusion stages

S-curve model: Integration of Milroy's and Rogers' model of diffusion stages (Kerremans 2015, 65)



Early adoption: Innovative users introduce new words.

Acceleration: Rapid spread through social networks.

Deceleration: Slower growth as market saturates.

Stabilisation: Word becomes established in conventional usage.

EC-Model: Entrenchment and conventionalisation

The dynamics of the linguistic system (Schmid 2020)

Conventionalisation: part of the conventional language system.

Entrenchment: storage in speakers' mental lexicon.

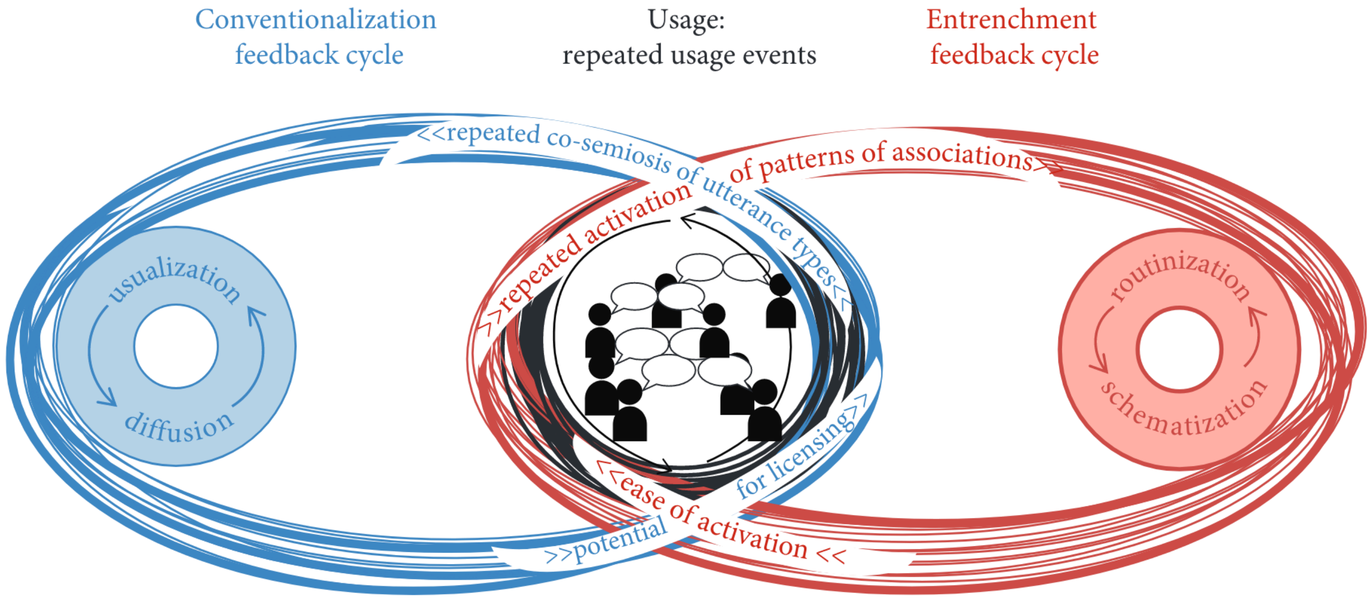
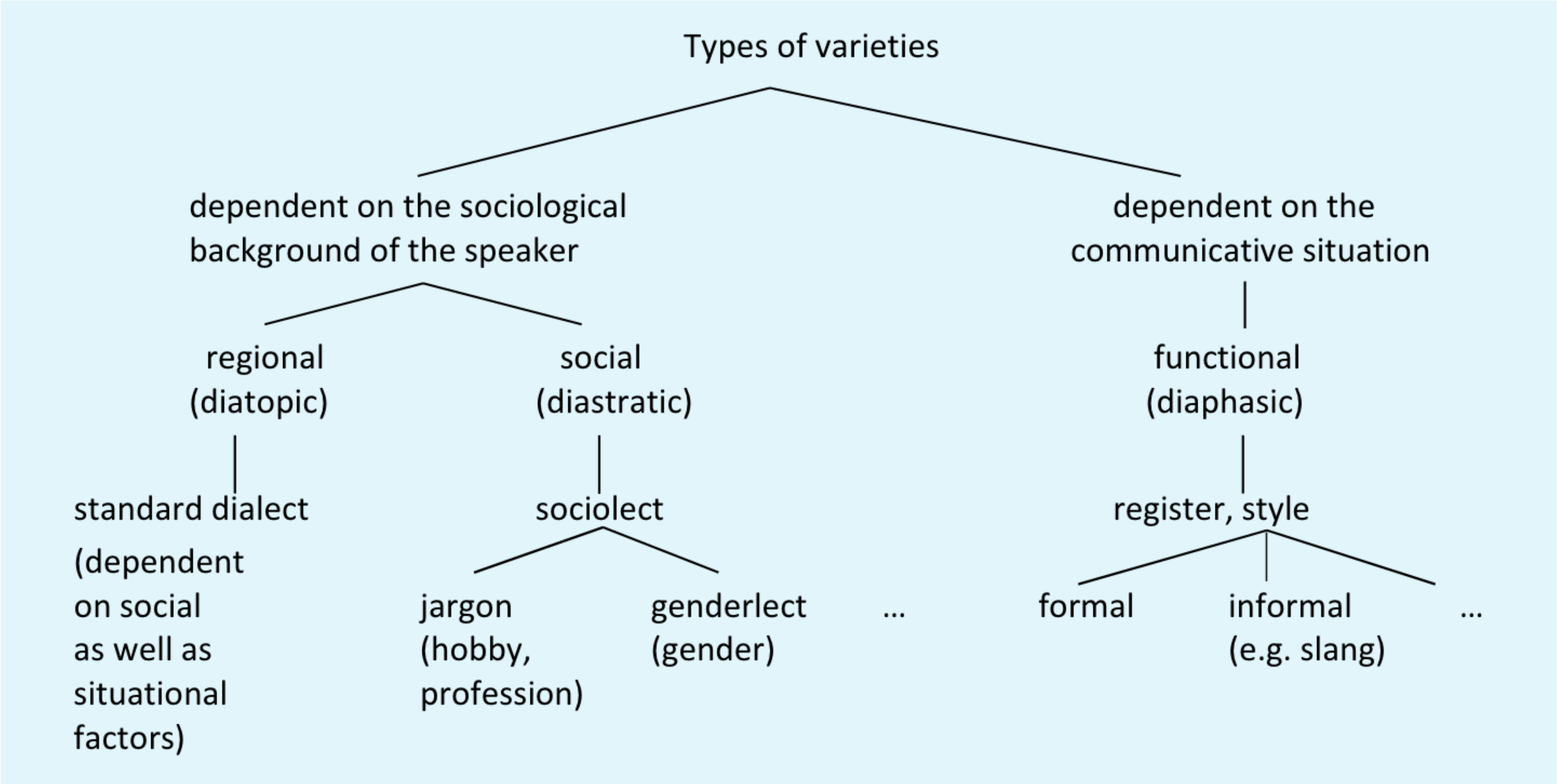


Figure 1.2 Language as a simple Tinguely machine

The interplay between usage, entrenchment, and conventionalisation (Schmid 2020, 4).

Pathways of diffusion

Types of linguistic variation and diffusion



Types of linguistic variation: diatopic, diastratic, and diaphasic (Kortmann 2020, 204)

Dimensions of diffusion

Across speakers and communities

- Regional spread
- Social group adoption
- Age group patterns

Across text types

- Formal vs informal contexts
- Academic vs popular writing
- Spoken vs written language

Examples

usage contexts	low	*hypostatization*	*electron*
	high	*alt-left*	*DNA*
		low	high
		speakers	

Operationalisation: frequency as indicator

Theoretical foundation

Frequency as indicator for entrenchment and conventionality (Stefanowitsch and Flach 2017):

- **Corpus-as-input**: language in corpora represents potential exposure to speakers.
- **Corpus-as-output**: language used by speakers represents degrees of entrenchment.

Frequency measures

Examples from Würschinger (2021):

Total frequency patterns

Most frequent:

TABLE 1 | Total usage frequency (FREQ) in the corpus. Most frequent lexemes.

Lexeme	**FREQ**
tweeter	7,367,174
fleek	3,412,807
bromance	2,662,767
twitterverse	1,486,873
blockchain	1,444,300
smartwatch	1,106,906

Around the median

TABLE 2 | Total usage frequency (FREQ) in the corpus. Examples around the median.

Lexeme	FREQ
white fragility	26,688
monthiversary	23,607
helicopter parenting	26,393
deepfake	20,101
newsjacking	20,930
twittosphere	20,035

Least frequent

TABLE 3 | Total usage frequency (FREQ) in the corpus. Least frequent lexemes.

Lexeme	**FREQ**
microflat	426
dogfishing	399
begpacker	283
halfalogue	245
rapugee	182
bediquette	164

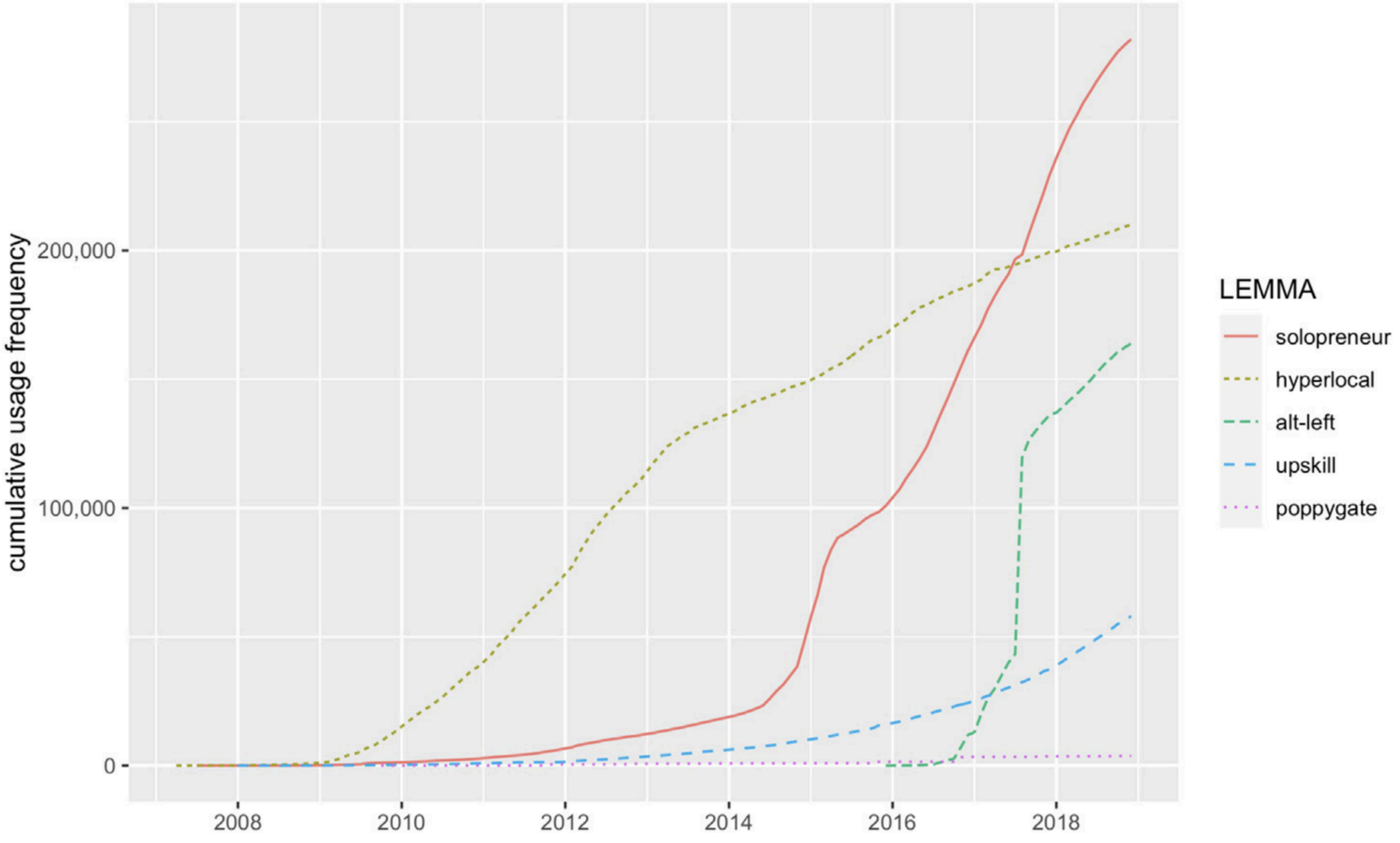
Case study selection

TABLE 4 | Total usage frequency (FREQ) in the corpus. Case study selection.

Lexeme	FREQ
alt-right	1,012,150
solopreneur	282,026
hyperlocal	209,937
alt-left	167,124
upskill	57,941
poppygate	3,807

Frequency over time (diachronic analysis)

Cumulative frequency



Raw frequency

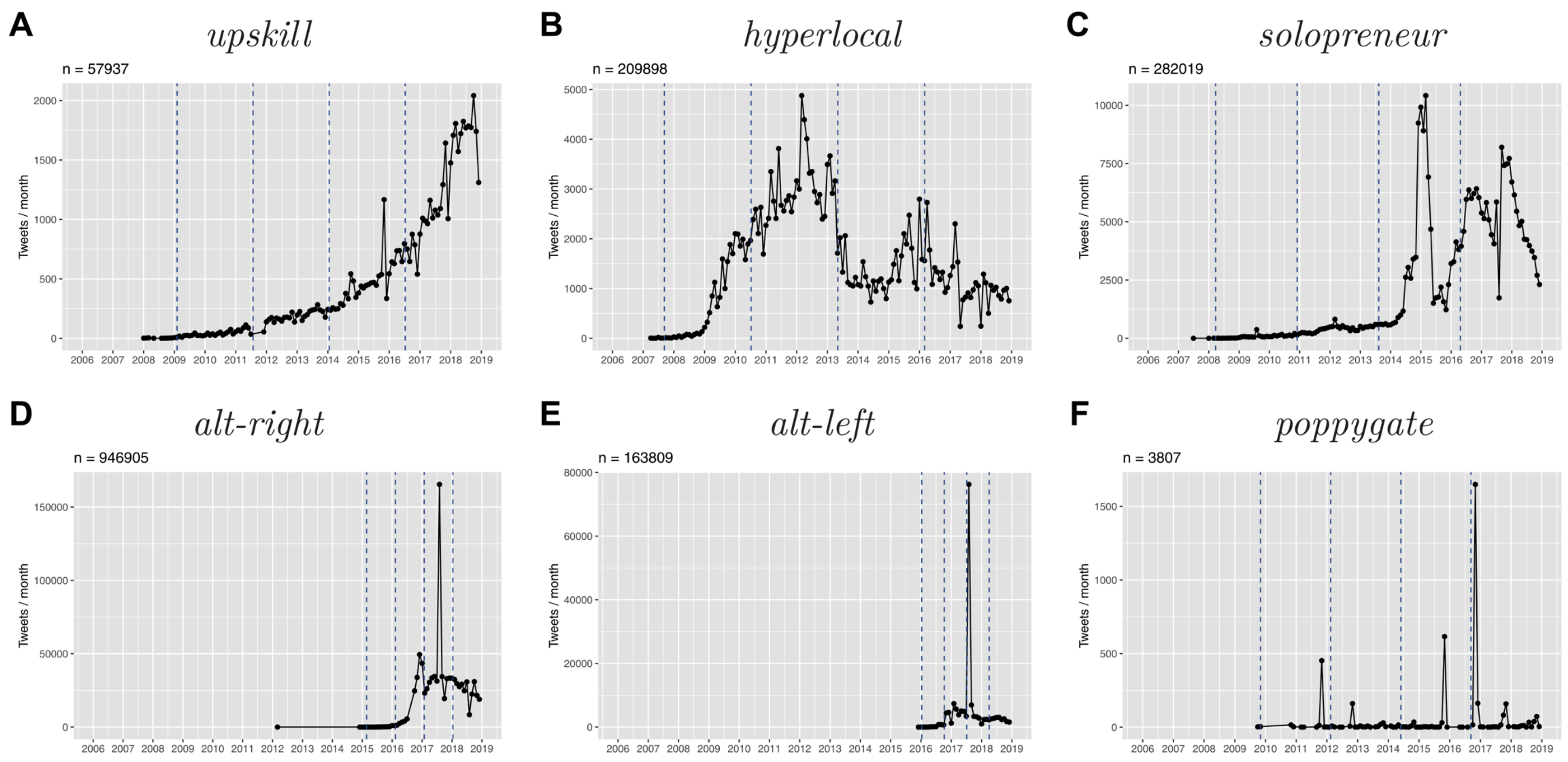


FIGURE 2 | Temporal dynamics in usage frequency for the selected neologisms.

Diffusion over time

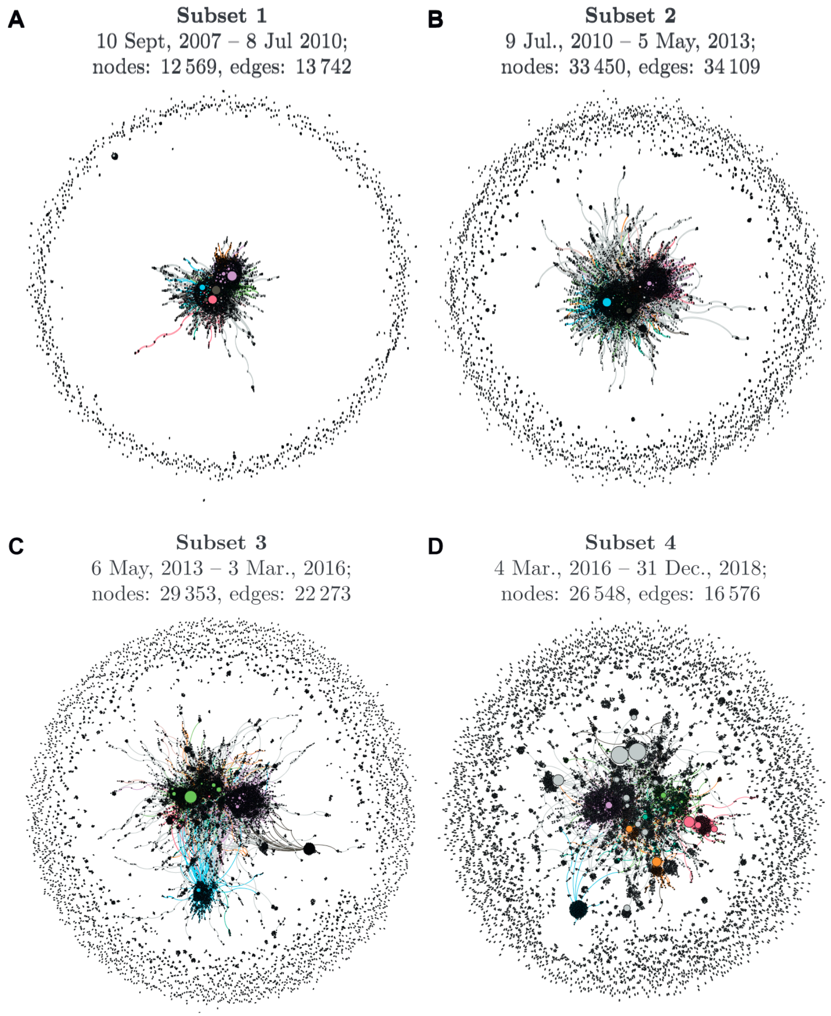
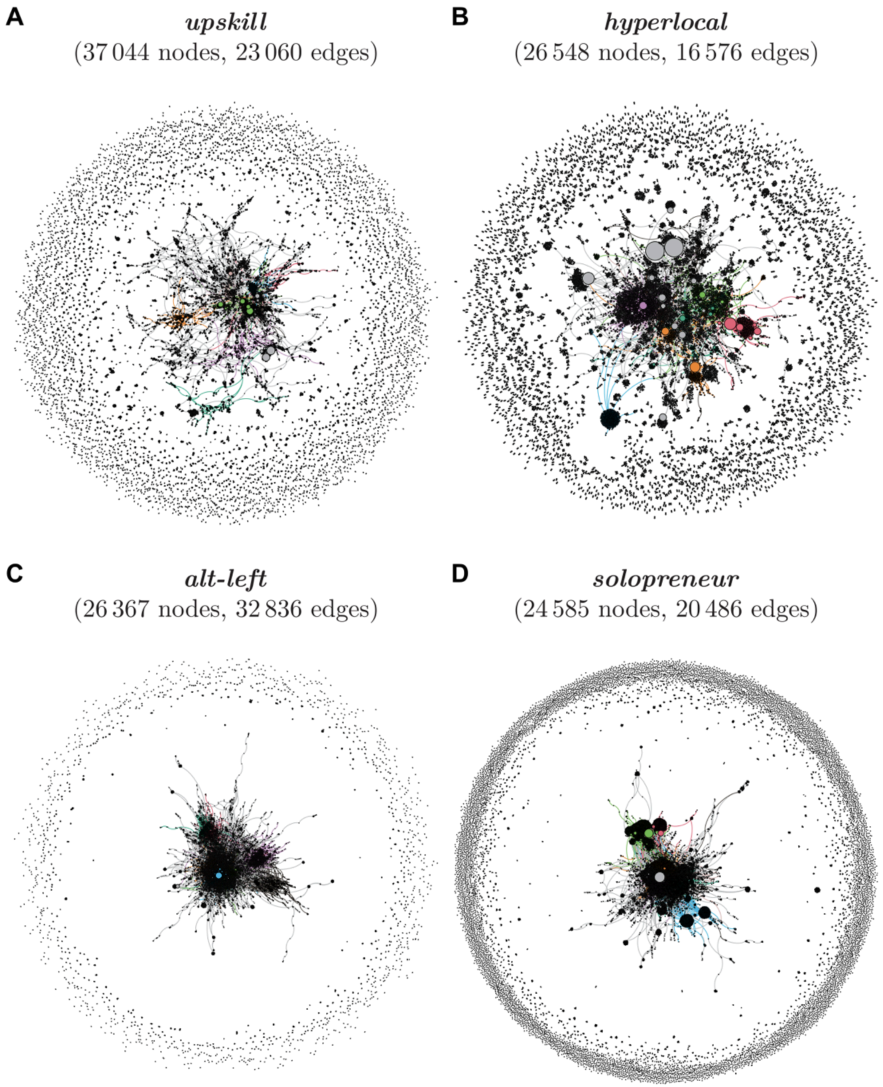


FIGURE 5 | Social network of diffusion for *hyperlocal* over time.

Diffusion across communities



Diffusion across text types

bro in COCA

SECTION	ALL	BLOG	WEB	TV/M	SPOK	FIC	MAG	NEWS	ACAD
FREQ	7272	453	328	5610	204	421	160	89	7
WORDS (M)	993	128.6	124.3	128.1	126.1	118.3	126.1	121.7	119.8
PER MIL	7.32	3.52	2.64	43.80	1.62	3.56	1.27	0.73	0.06
SEE ALL SUB-SECTIONS AT ONCE									

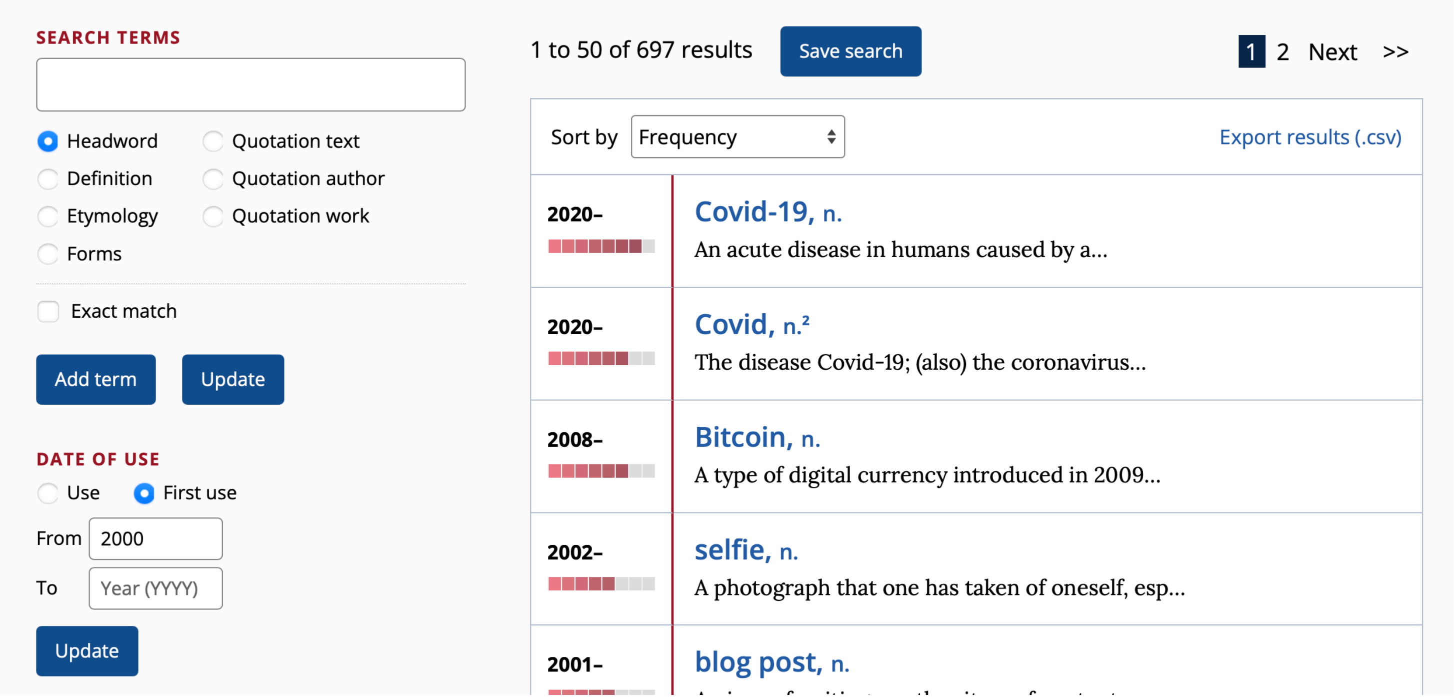
brother in COCA

SECTION	ALL	BLOG	WEB	TV/M	SPOK	FIC	MAG	NEWS	ACAD
FREQ	118471	7787	10361	35309	12205	26194	10489	12270	3856
WORDS (M)	993	128.6	124.3	128.1	126.1	118.3	126.1	121.7	119.8
PER MIL	119.30	60.55	83.39	275.69	96.76	221.38	83.19	100.79	32.19
SEE ALL SUB-SECTIONS AT ONCE									

Practice: identifying pattern of lexical innovation

OED's Advanced Search

Use the OED's Advanced Search to retrieve all words that have first been used since 2000.



Export the results as a `csv` file.

Excel analysis

Model sheet:
https://1drv.ms/x/c/9a2ec97d593520f9/EbTMdPaevpFJqGqbQT43l3YBf2Pg02PJ8On

1. import the `csv` file into Excel
2. create a `Table` that spans the imported data (without metadata)
3. created `Pivot Table`s to analyse the data

Analysis:

1. What are the most frequent word classes among the target neologisms?
2. What is their distribution across dates of first use?
3. Which word-formation processes are most common?
4. Which subject areas are most prevalent?

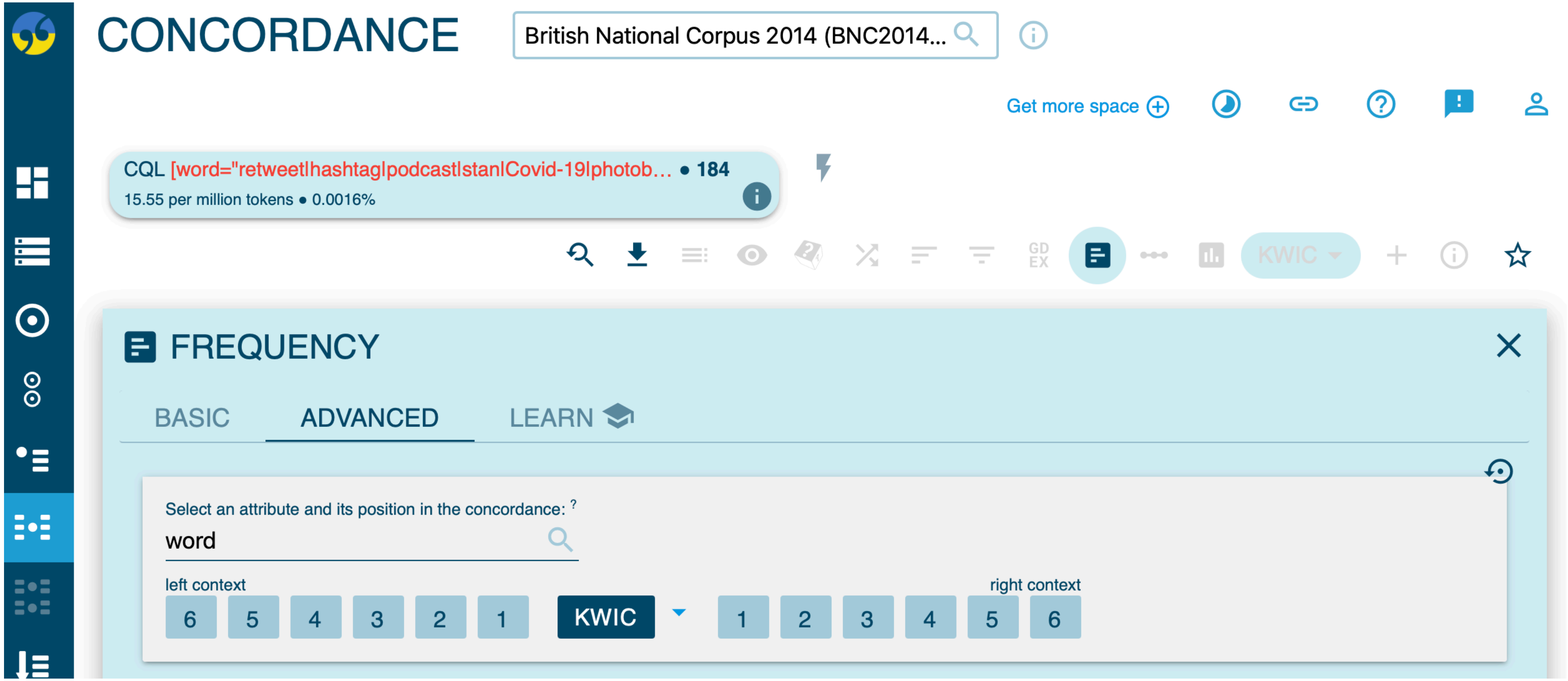
Sketch Engine analysis

Use Sketch Engine and the BNC 2014 Spoken to analyse the frequency and distribution of the target neologisms.

1. In Excel, selected a sample (e.g. `n=20`) of the target neologisms based on frequency (column `Frequency Band Number`).
2. In Sketch Engine, search for all of these neologisms in one go by concatenating them in a single CQL query. For example:

```
[word="retweet|hashtag|podcast|stan|Covid-
19|photobomb|upvote|Covid|deadname|Bitcoin|stablecoin|selfie|SARS|open-world|vlog|vape"]
```

Analyse the results by using the `Frequency` tool.



1. Frequency by `word`:

(8 items, 184 total frequency)

		Word	Frequency	Relative ?	% Of conc. ?
1	☐	selfie	65	5.49	35.33 %
2	☐	hashtag	54	4.56	29.35 %
3	☐	podcast	39	3.30	21.20 %
4	☐	retweet	10	0.85	5.43 %
5	☐	SARS	6	0.51	3.26 %
6	☐	stan	4	0.34	2.17 %
7	☐	vlog	3	0.25	1.63 %
8	☐	vape	3	0.25	1.63 %

2. Frequency by `Age range`:

(9 items, 184 total frequency)

		Age range ↓	Frequency	Relative density ?		
1	☐	===NONE===	1	79.74 %		•••
2	☐	0-10	4	168.93 %		•••
3	☐	11-18	52	463.15 %		•••
4	☐	19-29	57	84.17 %		•••
5	☐	30-39	20	74.92 %		•••
6	☐	40-49	17	65.20 %		•••
7	☐	50-59	23	123.50 %		•••
8	☐	60-69	7	41.01 %		•••
9	☐	70-79	3	32.44 %		•••

Summary

1. **Cultural and linguistic innovation are intertwined**: societal changes drive lexical innovation.
2. **S-curve model** describes diffusion patterns from innovation to stabilisation.
3. **EC-Model** explains entrenchment and conventionalisation through frequency.
4. **Diffusion occurs across multiple dimensions**: speakers, communities, and text types.
5. **Corpus methods enable empirical study** of innovation and diffusion patterns.

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